

PrimeCore Backup Server Technical Artefact

Implementation and verified recovery of the HQ DC archive

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Purpose and verified outcome

This artefact documents how HQ-BACKUP was connected, configured and tested in EVE-NG. It explains the commands and their effects, records unsuccessful attempts and corrections, and provides the complete screenshot evidence for review and repeatable maintenance.

The implemented result is a manual backup process: Samba creates an online Active Directory archive on HQ-DC, SCP transfers it to HQ-BACKUP, and Restic stores and restores it from an encrypted repository. Snapshot e709d068 restored an archive identical to the transferred original. An earlier local file test also preserved file contents, ownership and mode 640.

This is verified archive recovery, not a demonstrated recovery of a running domain. Automated generation and collection, retention, failure alerts, a separately protected off-host copy, and isolated domain recovery remain outstanding.

Evidence and attribution

Adam carried out the lab commands and supplied the screenshots. AI assistance supported troubleshooting, command explanations and preparation of this artefact. The online backup output identifies the domain as primecore.test. The screenshots do not independently establish which domain administrator username was used.

The evidence appendix embeds all 51 screenshots supplied in this conversation, including the two preceding DHCP incident screenshots. Evidence IDs B01 to B51 are used throughout. Original PNGs and SHA256 hashes are retained in evidence/Backup_Server. Commands in this document are explanatory records; destructive operations must not be rerun on disks already containing backups.

Reading order

Pages following this overview cover architecture, switch configuration, Linux networking, firewall access, storage, Restic validation, Samba backup, secure transfer, results and remaining work. The appendix follows the original screenshot sequence and explains what each image proves. It retains error screenshots as troubleshooting evidence.

Architecture and implemented configuration

Component	Observed setting
HQ-BACKUP	EVE node 40; Ubuntu 26.04 LTS; 2 vCPUs; 4096 MB RAM; 1 NIC
Backup network	ens3 at 192.168.91.10/24; gateway 192.168.91.1
Switch connection	HQ-BACKUP e0 to HQ-SW-OVS e11; Linux ens14; br0 access VLAN 91
Firewall uplink	HQ-SW-OVS ens4 trunk already includes VLAN 91
DNS and protected DC	HQ-DC 192.168.90.10; Samba 4.23.6; primecore.test
Disks	vda 30 GB OS; vdb 100 GB backup filesystem
Backup mount	/srv/backups; ext4; label PRIMECORE_BACKUP
Repository	/srv/backups/primecore; Restic 0.18.1; repository format 2
Tools	rsync 3.4.1 installed; demonstrated transfer uses SCP

Data flow

HQ-DC creates an application-aware Samba archive locally. HQ-BACKUP initiates SSH/SCP across pfSense from VLAN 91 to the server VLAN. The downloaded archive is placed in a protected staging directory and then backed up into Restic on vdb. Verification restores to a separate directory on HQ-BACKUP and compares the bytes.

Why isolate the backup server

The preceding Site A incident exposed lab DHCP to the home network (B01–B02). Jared was reported to have changed settings that appeared to mitigate it, but the exact change and cause were not established. HQ-BACKUP therefore uses an internal VLAN access port, with narrowly scoped routed services. A separate backup VLAN does not itself provide immutable storage or protection from EVE host loss.

No direct Cloud0 attachment was used for this server. The backup and source machines still depend on the same lab host; the second virtual disk is storage separation within that host, not an independent failure domain.

Switch attachment and VLAN reasoning

Evidence B03–B08 shows the disconnected server, exhausted adapter list, added e11, existing switch configuration and completed access-port assignment.

```
sudo ovs-vsctl show
ip -br link
```

ovs-vsctl show reads the Open vSwitch database to identify bridges, access tags and trunk VLAN lists. ip -br link lists Linux interfaces and link states in a compact form. Reading both prevented confusing EVE e11 with Linux ens11. The new adapter mapped to ens14; existing connections were preserved.

All original e0–e10 adapters were occupied. The switch was stopped and its adapter count increased from 11 to 12 through EVE, temporarily interrupting HQ connectivity. The new e11 was connected to HQ-BACKUP e0.

```
sudo ovs-vsctl --may-exist add-port br0 ens14 -- \
    set Port ens14 vlan_mode=access tag=91
sudo ovs-vsctl list Port ens14
```

add-port attaches ens14 to br0. --may-exist tolerates repeating the attachment when the port is already present on that bridge. The second operation sets access mode and tag 91, so Ubuntu can send untagged traffic while the switch assigns it to VLAN 91. The listing confirms those stored values. The existing ens4 trunk already allowed 91, so its VLAN list did not need changing.

Persistence and verification limits

Open vSwitch stores this change in its database. A later reboot test should confirm that image startup scripts do not overwrite it. The screenshot verifies configuration and the subsequent successful gateway ping verifies the tested path. Custom OpenFlow behaviour and every possible inter-VLAN flow were not independently audited.

Linux identity and network configuration

```
sudo hostnamectl set-hostname HQ-BACKUP
```

```
sudo nano /etc/hosts
```

hostnamectl changes the operating-system hostname. Updating the old srv-lnx-01 entry in /etc/hosts keeps local hostname resolution consistent. EVE node naming is a separate label. The resulting HQ-BACKUP prompt is visible, although the rename commands themselves were not captured.

```
ip -br addr
```

```
ip route
```

```
ls /etc/netplan
```

```
sudo cat /etc/netplan/*.yaml
```

These commands identify addresses, the routing table and the existing Netplan configuration before editing. The cloned configuration retained an old MAC match; the final file selects ens3 by interface name instead (B09).

```
sudo cp -n /etc/netplan/00-installer-config.yaml \
/etc/netplan/00-installer-config.yaml.bak
```

```
sudo nano /etc/netplan/00-installer-config.yaml
```

The copy preserves the previous file without replacing an existing .bak. The recorded cp portability warning is not proof of a failed copy, but backup existence was not separately shown. nano edits the YAML. The final accepted configuration is below (B24–B25).

```
network:
  version: 2
  ethernet:
    ens3:
      dhcp4: false
      dhcp6: false
      addresses:
        - 192.168.91.10/24
      nameservers:
        addresses:
          - 192.168.90.10
      routes:
        - to: default
          via: 192.168.91.1
```

dhcp4 and dhcp6 disable DHCP address acquisition. They do not disable all IPv6 behaviour. The address includes a /24 prefix; the route supplies the IPv4 gateway; nameservers configures the client resolver. Availability of .10 and exclusion from a DHCP allocation pool should be retained in the address register; a dedicated exclusion screenshot was not supplied.

Network validation and troubleshooting

```
sudo chmod 600 /etc/netplan/00-installer-config.yaml
sudo netplan generate
sudo netplan try
```

Mode 600 limits configuration access to root. generate parses the YAML and builds backend configuration without being the final apply step. try applies it temporarily and requests confirmation; the accepted prompt is recorded. Indentation errors and a root-level routes key were corrected before acceptance (B11–B14). A shell tee here-document was proposed to preserve indentation, but successful use of that alternative is not evidenced.

```
ip -br addr
ip route
ping -c 4 192.168.91.1
ip neigh show 192.168.91.1
```

The first two commands distinguish an address or routing problem from filtering. ping sends four bounded echo requests. ip neigh examines ARP neighbour resolution: a cached MAC with STALE state supports prior link-layer communication and is not an error. Initial pings failed despite a valid address, route and neighbour. After adding the specific ICMP rule, four replies and zero loss were recorded (B14–B17).

```
hostname -I
sudo ss -lntup 'sport = :53'
```

On HQ-DC these identify its address and listening DNS sockets. ss uses listening, numeric, TCP, UDP and process output; the filter limits the display to port 53. A listener alone does not prove client reachability.

```
resolvectl status ens3
resolvectl query ubuntu.com
```

On HQ-BACKUP these inspect resolver configuration and test an actual name lookup. The early failure said no appropriate name servers, so the missing client nameservers block was fixed rather than widening firewall access. The final output shows 192.168.90.10 and resolved external addresses. IPv6 answers in DNS do not establish usable IPv6 routing.

The resolvectl DefaultRoute field concerns resolver routing, not the presence of the IPv4 default route. clear or Ctrl+L only clears the terminal display and has no network effect.

Firewall access and software installation

Rules belong on HQ pfSense BACKUP_VLAN because HQ-BACKUP initiates these connections there. Source ports remain Any. Specific service ports are destination ports. The observed DNS edit was corrected from any-53 to 53-53. A single IP under Address or Alias is a host match; its disabled prefix field can remain blank.

Purpose	Source and destination	Allowed traffic
Gateway test	192.168.91.10 to 192.168.91.1	IPv4 ICMP echo request
DNS	192.168.91.10 to 192.168.90.10	IPv4 TCP and UDP 53
Temporary downloads	192.168.91.10 to Any	IPv4 TCP 80 only
DC archive transfer	192.168.91.10 to 192.168.90.10	IPv4 TCP 22

Pass rules were instructed above blocking rules with logging enabled and Save/Apply Changes. Rule screenshots exist for ICMP, DNS and HTTP. The SSH rule was instructed but no rule screenshot was supplied; successful SCP proves the observed SSH path, not the exact rule responsible. No broad backup-VLAN-to-any allowance is claimed.

```
sudo apt update
sudo apt -o Acquire::ForceIPv4=true update
sudo apt -o Acquire::ForceIPv4=true install restic rsync
restic version
rsync --version | head -n 1
```

apt update refreshes package metadata; it does not install the displayed upgrades. The first attempt failed: IPv6 lacked a route and IPv4 HTTP timed out. The temporary TCP 80 rule and ForceIPv4 option produced a successful index download. ForceIPv4 applies to that invocation and does not repair a blocked IPv4 path. install requests the backup tools; version output verifies Restic 0.18.1 and rsync 3.4.1. The install transaction itself is not shown, and rsync was not used for the demonstrated transfer.

The temporary HTTP rule allows port 80 to any destination from one server. Removal or replacement with an approved update policy is outstanding; do not describe it as already disabled. The available 173 package upgrades were not demonstrated as installed.

Adding the backup disk on EVE

Evidence B30–B36 distinguishes disk creation from disk attachment. The node editor shows no direct disk control. EVE host commands were run as root, not inside the guest.

```
df -h /opt/unetlab
```

```
sudo find /opt/unetlab/tmp -type d -name 40
```

df checks actual host free space before allocating storage; 295 GB was available at that time. find locates candidate node 40 directories. Node IDs are not globally unique across labs, so the lab context and existing system disk were checked before creation.

```
cd /opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40
```

```
ls -lh
```

```
cat wrapper.txt
```

The listing identifies virtioa.qcow2 as the existing OS image. The wrapper log records the node process stopping.

sudo poweroff inside HQ-BACKUP provides a clean shutdown before changing storage attachment.

```
if [ ! -e virtiob.qcow2 ]; then
```

```
  /opt/qemu/bin/qemu-img create -f qcow2 virtiob.qcow2 100G
```

```
fi
```

```
chown root:unl virtiob.qcow2
```

```
chmod 644 virtiob.qcow2
```

The existence guard avoids recreating an existing disk. qemu-img creates a sparse QCOW2 disk with a 100 GB virtual capacity; real consumption grows with use. chown and chmod match the existing node disk access pattern. An earlier command omitted virtiob.qcow2 and failed because 100G was interpreted as the filename. Do not rerun unguarded creation over an existing backup disk.

The guest still lacked vdb after creation. The following was appended to the existing QEMU custom options while the node was stopped:

```
-drive file=virtiob.qcow2,if=virtio,format=qcow2
```

This explicitly attaches the disk file using VirtIO. The later guest listing shows the 100 GB vdb. The option-edit screen itself was not captured. The file remains in EVE node working storage: wiping the node can delete it, and a lab export must not be assumed to include it.

Formatting and persistent mounting

```
lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS  
cat /etc/os-release
```

lsblk distinguishes the 30 GB vda OS disk from the new blank 100 GB vdb. The OS file confirms Ubuntu 26.04 LTS. This identification is the safeguard before the one-time destructive formatting step.

```
sudo mkfs.ext4 -L PRIMECORE_BACKUP /dev/vdb  
sudo mkdir -p /srv/backups  
sudo mount /dev/vdb /srv/backups  
sudo blkid /dev/vdb
```

mkfs.ext4 creates the filesystem and label on the newly created vdb only. It must not be repeated on populated disks. mkdir -p makes the mount directory, mount attaches the filesystem, and blkid reads its UUID. A whole-disk filesystem is used here; no vdb1 partition was created.

The following entry was added to /etc/fstab without replacing the OS or swap entries:

```
UUID=4ee001f1-8953-40b4-a3cf-20173e67a8ed /srv/backups ext4 defaults 0 2
```

Enter this as one physical line in fstab. UUID is a stable identifier if device names change. defaults supplies normal mount options, 0 disables the legacy dump flag, and 2 schedules filesystem checking after the root filesystem. The line begins UUID=, not /UUID=.

```
sudo systemctl daemon-reload  
sudo findmnt --verify  
df -h /srv/backups
```

daemon-reload refreshes systemd configuration after the fstab edit. findmnt validates the mount table; zero parse errors and zero errors were reported. Its warning concerns the pre-existing swap file. df confirms the actual mounted filesystem, showing about 98 GB total and 93 GB available after filesystem overhead and reserved space. These checks do not replace a reboot persistence test, which remains outstanding.

Encrypted repository and local restore test

```
sudo chmod 700 /srv/backups
sudo mountpoint -q /srv/backups && \
  sudo restic -r /srv/backups/primecore init
sudo restic -r /srv/backups/primecore check
```

Mode 700 restricts the mounted backup directory to root. mountpoint -q verifies that storage is mounted, and && prevents initialisation when that test fails, avoiding accidental use of the OS disk. Restic init creates an encrypted repository protected by the prompted password. The password must be held separately for recovery and is not recorded here. The initial check passed but had zero snapshots; it was not evidence that business data had been backed up.

```
mkdir -p ~/primecore-backup-test
printf 'PrimeCore backup and restore test\n' > ~/primecore-backup-test/test.txt
chmod 640 ~/primecore-backup-test/test.txt
```

A small known-content file gives a controlled restore test without deleting business data. Mode 640 allows owner read/write, group read, and no access to others.

```
sudo mountpoint -q /srv/backups && \
  sudo restic -r /srv/backups/primecore backup \
    /home/labadmin/primecore-backup-test --tag restore-test
sudo restic -r /srv/backups/primecore restore latest \
  --tag restore-test --target /var/tmp/primecore-restore-test
```

backup records file data and metadata; the tag distinguishes this validation from DC archive jobs. restore latest selects the latest matching tagged snapshot and writes to a separate target rather than overwriting the source. Absolute source paths reappear beneath the restore target.

```
sudo diff -s /home/labadmin/primecore-backup-test/test.txt \
  /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt
sudo stat -c '%U:%G %a %n' \
  /home/labadmin/primecore-backup-test/test.txt \
  /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt
```

diff -s explicitly reports identical byte content. stat prints owner, group, numeric permissions and pathname.

Snapshot 157bfc4c passed both comparisons, preserving labadmin:labadmin and 640 (B41). This tests basic Unix mode preservation, not every ACL, extended attribute or application recovery requirement.

Creating an application aware domain backup

HQ-MONITOR was deferred at Adam's request because it appeared occupied. HQ-DC was selected instead. A supported Samba online backup captures the directory and SYSVOL without treating a live database-file copy as a valid recovery method.

```
sudo samba-tool --version
sudo testparm -s --parameter-name='server role'
sudo samba-tool domain backup online --help
sudo ss -lntp 'sport = :22'
sudo samba-tool group listmembers "Domain Admins"
```

The version and role checks establish Samba 4.23.6 and an AD domain controller. --help checks locally available options. ss confirms a listening SSH port. The group command was supplied to identify domain administrator accounts, but its output was not captured. Linux labadmin and the domain Administrator are different identities. The testparm weak-crypto message describes compatibility settings; it is not a backup failure or a completed security review.

```
sudo install -d -m 700 /var/backups/primecore-ad
sudo samba-tool domain backup online --server=192.168.90.10 \
--targetdir=/var/backups/primecore-ad -U Administrator
sudo ls -lh /var/backups/primecore-ad
```

install -d creates a root-restricted staging directory. online requests a supported running-domain backup; --server chooses HQ-DC; --targetdir directs the archive; -U selects the authorised domain account and prompts for its password. Administrator was the example account, not independently evidenced as the account actually used. No --no-secrets or --no-sysvol exclusion was requested.

The output records replication of directory partitions and DNS zones, SYSVOL backup and creation of the following archive (B45–B46):

```
samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2
```

The archive listing is about 1.4 MB. Temporary provisioning output mentions a generated krb5.conf under the backup work directory. That diagnostic message must not be treated as a request to replace the live DC configuration.

The archive filename records the DC clock without an explicit timezone. Do not silently convert that filename to AEST. The later Restic restore output explicitly states UTC and can be converted reliably.

Validating the Samba archive

```
sudo find /var/backups/primecore-ad -maxdepth 1 -type f \
-name '*.tar.bz2' -exec chmod 600 {} +
```

find selects only top-level regular backup archives, and chmod 600 removes group and other read access. The parent directory was already mode 700. This tightens file permissions without making the DC directory publicly accessible.

The attempted tar -tjf command lists a bzip2-compressed tar archive: t lists, j selects bzip2, and f names the file. Redirecting the list to /dev/null avoids a large terminal listing, while && prints success only on exit status zero. It failed because the external bzip2 executable was absent. No corruption conclusion followed from that error.

Python was used as an available alternative, with the following logic. The two-line path expression below is equivalent to the captured full pathname:

```
sudo python3 - <<'PY'
import bz2
import tarfile
from pathlib import Path

p = Path('/var/backups/primecore-ad') / (
    'samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2')

with bz2.open(p, 'rb') as archive:
    while archive.read(1024 * 1024):
        pass
print('Compressed data check passed')

with tarfile.open(p, 'r:bz2') as archive:
    print(f'Archive readable: {len(archive.getmembers())} entries')
PY
```

The quoted here-document marker passes literal Python source without shell expansion. bz2 reads the entire compressed stream in 1 MiB chunks, avoiding loading the entire decompressed backup into memory. tarfile opens the archive using built-in bzip2 support and enumerates its members without extracting them. Both checks completed, reporting 30 entries (B48).

Readability and decompression are necessary checks, but do not establish AD consistency after restoration, usable authentication, working DNS or correct Group Policy. Those require a separate isolated recovery exercise.

Secure transfer and identity verification

The demonstrated transfer uses the existing Linux labadmin account and interactive authentication. It does not implement the proposed future dedicated backup service account or unattended key authentication.

```
sudo install -d -o labadmin -m 700 /home/labadmin/backup-export
sudo install -o labadmin -m 600 \
  /var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2 \
  /home/labadmin/backup-export/hq-dc-ad.tar.bz2
sha256sum /home/labadmin/backup-export/hq-dc-ad.tar.bz2
sudo ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
```

On HQ-DC, install copies the archive into a restricted export directory while setting owner and permissions. This avoids enabling root SSH or opening the root-only archive directory. The checksum identifies file bytes; ssh-keygen displays the public host-key fingerprint to authenticate the destination independently of the first SSH prompt. Its root@srv-lnx-01 comment is historical metadata, not proof of a wrong host. Cloned-image SSH key uniqueness still needs review.

```
mkdir -p ~/backup-staging/hq-dc
chmod 700 ~/backup-staging ~/backup-staging/hq-dc
scp -o HostKeyAlgorithms=ssh-ed25519 \
  labadmin@192.168.90.10:/home/labadmin/backup-export/hq-dc-ad.tar.bz2 \
  ~/backup-staging/hq-dc/
sha256sum ~/backup-staging/hq-dc/hq-dc-ad.tar.bz2
```

On HQ-BACKUP, the directories restrict access to the staging copy. SCP encrypts the transfer over SSH, and the algorithm option makes the displayed host key directly comparable to the ED25519 fingerprint read on HQ-DC. The known-hosts acceptance and completed transfer are recorded. The earlier concatenated chmod/scp entry was corrected by running separate commands (B49–B50).

Both endpoints reported the same SHA256 checksum:

```
b9c6e8116f1b7fff0a2e5aa3732444d70c5d58744574d9a2ec3ac6e51a32234f
```

The export and staging copies are access-restricted plaintext archives outside Restic encryption. Their cleanup and retention need an explicit operational policy; no deletion of these copies is claimed.

Protecting and recovering the domain archive

```
sudo mountpoint -q /srv/backups && \  
  sudo restic -r /srv/backups/primecore backup \  
    /home/labadmin/backup-staging/hq-dc --tag hq-dc-ad \  
sudo restic -r /srv/backups/primecore restore latest \  
  --tag hq-dc-ad --target /var/tmp/hq-dc-restore-check \  
sudo diff -s /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 \  
  /var/tmp/hq-dc-restore-check/home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2
```

The mount guard prevents a backup job from proceeding when the intended filesystem is absent. The distinct hq-dc-ad tag separates domain archives from restore-test snapshots. Restoring to a separate path preserves the source. diff compares the recovered archive bytes with the transferred archive.

Evidence B51 records snapshot e709d068, about 1.394 MiB of source data, and a successful restore of five file/directory entries. Those entries include the directory hierarchy, not five independent AD backups. The final diff reports identical files.

Test	Observed result	Scope
Gateway	4 replies; 0 percent loss	HQ-BACKUP to 192.168.91.1
DNS	ubuntu.com resolved	Query through 192.168.90.10
Repository	Initial check found no errors	Empty repository at that time
Local restore	157bfc4c; identical; 640 retained	One controlled test file
Archive validation	Compressed read passed; 30 entries	Samba tar archive readability
Transfer	SCP complete; SHA256 identical	DC export to backup staging
Archive recovery	e709d068; identical restored archive	Restic round trip only

Snapshot e709d068 is explicitly dated 30 September 2026 04:42:30 UTC in the output, equivalent to 14:42:30 AEST. The local test snapshot is dated 29 September 2026 22:58:00 UTC, equivalent to 30 September 08:58:00 AEST. These timestamps document the jobs; they are not measured recovery-time commitments.

Operational limits and next implementation work

What is complete

The server, access VLAN, static address, DNS client and tested service paths are configured. A second disk is formatted and mounted with an fstab entry. Restic has an encrypted repository, a verified local-file restore, and a verified round trip of a supported Samba AD archive. These are manual demonstrated outcomes.

What remains

Automate fresh Samba backup generation before every scheduled collection; repeatedly backing up the same staged archive does not provide a fresh recovery point. Use a dedicated restricted identity and scoped SSH key or supported authentication design, rather than storing a domain administrator password in a script. Confirm timestamps and job dependencies.

Implement retention and failure monitoring. The existing draft policy proposes daily AD backups, 30 daily and 12 monthly recovery points, a 24-hour AD recovery point objective and an 8-hour recovery time objective. These are proposed targets, not measured achievements. Monitor job success, age of the newest usable archive, repository space and independent-copy status.

Keep an independent offline or immutable copy outside this EVE host with separately controlled recovery credentials. A second QCOW2 disk under the same node working directory is not off-host protection. Record how node wipe, lab export and host failure affect both disks, and preserve VM definitions and storage separately.

Test recovery of the domain in an isolated environment using the supported Samba restore workflow. Validate identity, DNS, SYSVOL, authentication and policy before claiming domain recovery. Never introduce a recovered clone into the live domain for an uncontrolled test.

Verify mount and switch configuration persistence after a controlled reboot. Remove or replace the temporary HTTP rule, review cloned SSH host-key uniqueness, and determine retention or secure disposal of unencrypted staging and restore-test copies. Review detailed permissions and any required ACLs beyond the basic local file-mode test.

Recovery and change safety

Never repeat mkfs.ext4 against the populated backup disk. Preserve the current Netplan and fstab files before future edits. Disabling a temporary firewall rule is preferable to widening unrelated access. Keep the Restic password recoverable outside the VM. This report contains paths, configuration and evidence, but no passwords or private keys.

Technical references and evidence guide

Primary references explain the commands and design. Local version output and supplied results remain the authority for what actually ran in this lab.

Open vSwitch access VLAN configuration

<https://docs.openvswitch.org/en/latest/howto/vlan/>

Netplan configuration examples

<https://github.com/canonical/netplan/blob/main/doc/examples.md>

Netplan temporary application and rollback

<https://github.com/canonical/netplan/blob/main/doc/netplan-try.md>

EVE QEMU disk naming

<https://www.eve-ng.net/index.php/documentation/qemu-image-namings/>

EVE node working directory layout

<https://www.eve-ng.net/index.php/documentation/howtos/howto-save-your-settings-to-be-as-default-on-qemu-node/>

Restic repositories and passwords

https://restic.readthedocs.io/en/stable/030_preparing_a_new_repo.html

Restic restoring backups

https://restic.readthedocs.io/en/stable/050_restore.html

Samba tool reference

<https://www.samba.org/samba/docs/current/man-html/samba-tool.8.html>

Project companions: HQ_Network_Addressing_Artefact.md;

PrimeCore_Technical_Artefacts/06_PrimeCore_Backup_Policy.docx; Security_Incident_Register.md, INC-002; Final_Report_Notes.md. Policy targets are distinguished from implemented controls above.

The following appendix includes all 51 original screenshots in sequence. B01–B02 are incident context; B03–B17 cover VLAN and addressing; B18–B29 cover DNS and packages; B30–B39 cover storage; B40–B41 cover Restic and local recovery; B42–B51 cover Samba backup, transfer and archive recovery. Each figure retains its original resolution within the editable document and has an evidence interpretation.

Evidence B01 Site A interface map

Context for isolation: SITEA_EMP_WIFI is 192.168.111.1 on VLAN 11. SITEA_MGMT is the renamed LAN. This is not the HQ backup gateway.

```
WAN (wan)          -> vtnet0      -> v4: 172.16.101.2/30
SITEA_MGMT (lan)    -> vtnet1.99 -> v4: 192.168.199.1/24
SITEA_EMPLOYEE (opt1) -> vtnet1.10 -> v4: 192.168.110.1/24
SITEA_EMP_WIFI (opt2) -> vtnet1.11 -> v4: 192.168.111.1/24
SITEA_VISITOR (opt3) -> vtnet1.20 -> v4: 192.168.120.1/24
SITEA_PRINTER (opt4) -> vtnet1.30 -> v4: 192.168.130.1/24
SITEA_VIDEO (opt5)   -> vtnet1.40 -> v4: 192.168.140.1/24
SITEA_CCTV (opt6)    -> vtnet1.41 -> v4: 192.168.141.1/24
SITEA_IOT (opt7)     -> vtnet1.50 -> v4: 192.168.150.1/24
SITEA_SERVER (opt8)  -> vtnet1.90 -> v4: 192.168.190.1/24
SITEA_BACKUP (opt9)  -> vtnet1.91 -> v4: 192.168.191.1/24

0) Logout / Disconnect SSH      9) pfTop
1) Assign Interfaces            10) Filter Logs
2) Set interface(s) IP address 11) Restart GUI
3) Reset admin account and password 12) PHP shell + Netgate pfSense Plus tools
4) Reset to factory defaults    13) Update from console
5) Reboot system               14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell
```

Figure B01 Site A interface map

Evidence B02 Home devices in lab DHCP leases

The earlier incident shows home devices receiving lab leases. This motivated an internal switch access port for the backup server. The exact escape path remains unconfirmed.

Leases							
	IP Address	MAC Address	Hostname	Description	Start	End	Actions
✓↑	192.168.111.215	48:2a:e3:bd:33:08	Darth-Ally		2026/09/28 09:53:15	2026/09/28 11:53:15	⊕⊕
✓↑	192.168.111.225	30:34:db:33:ca:a7	LG_Smart_Laundry2_open		2026/09/28 09:52:35	2026/09/28 11:52:35	⊕⊕
✓↑	192.168.111.224	84:0d:8e:80:ea:80	ESP_80EA80		2026/09/28 09:52:30	2026/09/28 11:52:30	⊕⊕
✓↑	192.168.111.227	cc:50:e3:f1:d6:eb	ESP_F1D6EB		2026/09/28 09:52:30	2026/09/28 11:52:30	⊕⊕
✓↑	192.168.111.229	ec:fa:bc:72:c6:45	ESP_72C645		2026/09/28 09:52:24	2026/09/28 11:52:24	⊕⊕
✓↓	192.168.111.217	68:d7:9a:5d:a6:a5	ADHome		2026/09/28 09:41:00	2026/09/28 11:41:00	⊕⊕⏻🗑
✓↓	192.168.120.200	00:50:79:66:68:0f	VPCS		2026/09/28 09:23:52	2026/09/28 11:23:52	⊕⊕⏻🗑
✓↑	192.168.111.202	10:d5:61:f4:fb:be	wlan0		2026/09/28 09:23:00	2026/09/28 11:23:00	⊕⊕
✓↓	192.168.111.213	d8:b3:2f:d4:85:91	Darth-Ally		2026/09/28 09:22:14	2026/09/28 11:22:14	⊕⊕⏻🗑
✓↓	192.168.141.200	00:50:79:66:68:15	VPCS		2026/09/28 09:16:21	2026/09/28 11:16:21	⊕⊕⏻🗑
✓↑	192.168.111.210	fa:2f:31:81:24:48	Ashleigh-s-S24		2026/09/28 09:15:48	2026/09/28 11:15:48	⊕⊕
✓↑	192.168.110.200	00:50:00:00:1a:00	srv-lnx-01		2026/09/28 09:12:45	2026/09/28 11:12:45	⊕⊕
✓↓	192.168.190.202	00:50:00:00:18:00	srv-lnx-01		2026/09/28 09:12:41	2026/09/28 11:12:41	⊕⊕⏻🗑
✓↑	192.168.111.211	84:b8:b8:56:7b:95			2026/09/28 09:10:38	2026/09/28 11:10:38	⊕⊕
✓↑	192.168.140.200	00:50:00:00:1e:00	srv-lnx-01		2026/09/28 09:10:06	2026/09/28 11:10:06	⊕⊕
✓↓	192.168.111.234	7a:df:f1:df:3c:89	Adam-s-Z-Fold3		2026/09/28 09:08:51	2026/09/28 11:08:51	⊕⊕⏻🗑
✓↑	192.168.111.228	d0:73:d5:55:a2:21	LIFX-Mini-55A221		2026/09/28 09:08:32	2026/09/28 11:08:32	⊕⊕
✓↓	192.168.111.214	b4:c9:b9:25:12:05	dgiot		2026/09/28 09:08:07	2026/09/28 11:08:07	⊕⊕⏻🗑
✓↓	192.168.111.230	14:0a:02:b5:21:e3			2026/09/28 09:07:57	2026/09/28 11:07:57	⊕⊕⏻🗑
✓↑	192.168.111.233	d0:73:d5:55:b4:43	LIFX-Mini-55B443		2026/09/28 09:07:57	2026/09/28 11:07:57	⊕⊕
✓↑	192.168.111.231	1c:f2:9a:0d:e3:f6			2026/09/28 09:07:54	2026/09/28 11:07:54	⊕⊕
✓↑	192.168.111.226	38:77:07:4e:ac:ce	Smart_Camera		2026/09/28 09:07:48	2026/09/28 11:07:48	⊕⊕
✓↑	192.168.111.208	a6:d0:e4:66:55:60	Adam-s-S24-Ultra		2026/09/28 09:07:47	2026/09/28 11:07:47	⊕⊕
✓↓	192.168.111.212	7c:a7:b0:a9:8c:27	059553842		2026/09/28 09:07:47	2026/09/28 11:07:47	⊕⊕⏻🗑
✓↑	192.168.111.216	38:26:56:a0:8c:c6	Lounge-Room-TV		2026/09/28 09:06:50	2026/09/28 11:06:50	⊕⊕
✓↓	192.168.111.200	00:50:79:66:68:0e	VPCS		2026/09/28 09:02:23	2026/09/28 11:02:23	⊕⊕⏻🗑

Figure B02 Home devices in lab DHCP leases

HQ-BACKUP is disconnected below HQ. The intended path is through HQ-SW-OVS and HQ-FW, not a direct Cloud0 connection.



Evidence B04 All original switch adapters occupied

Adapters e0 through e10 already have endpoints. Reusing one would displace an existing service.

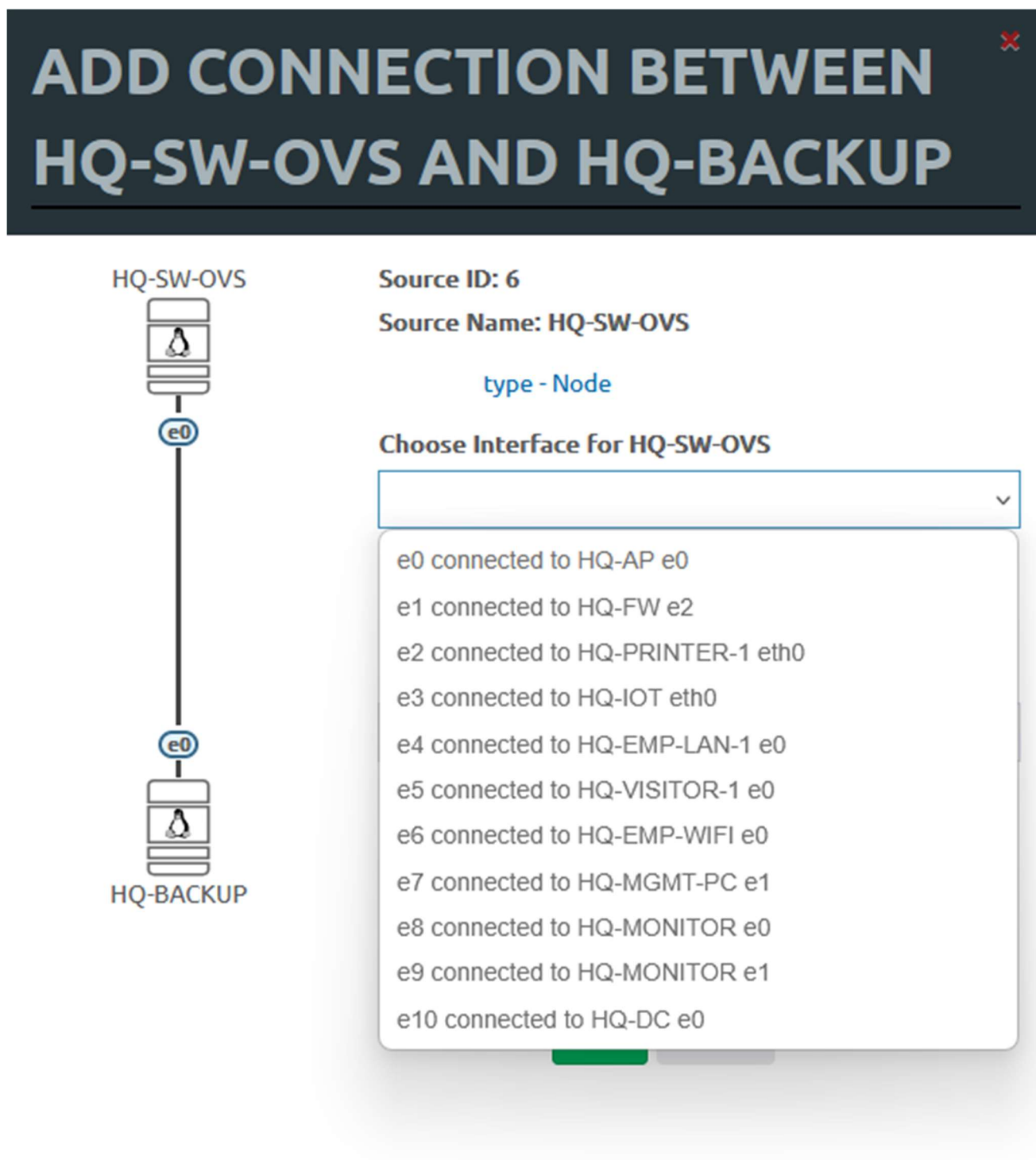


Figure B04 All original switch adapters occupied

Evidence B05 New switch adapter selected

e11 is available after increasing the adapter count. It is selected for HQ-BACKUP e0.

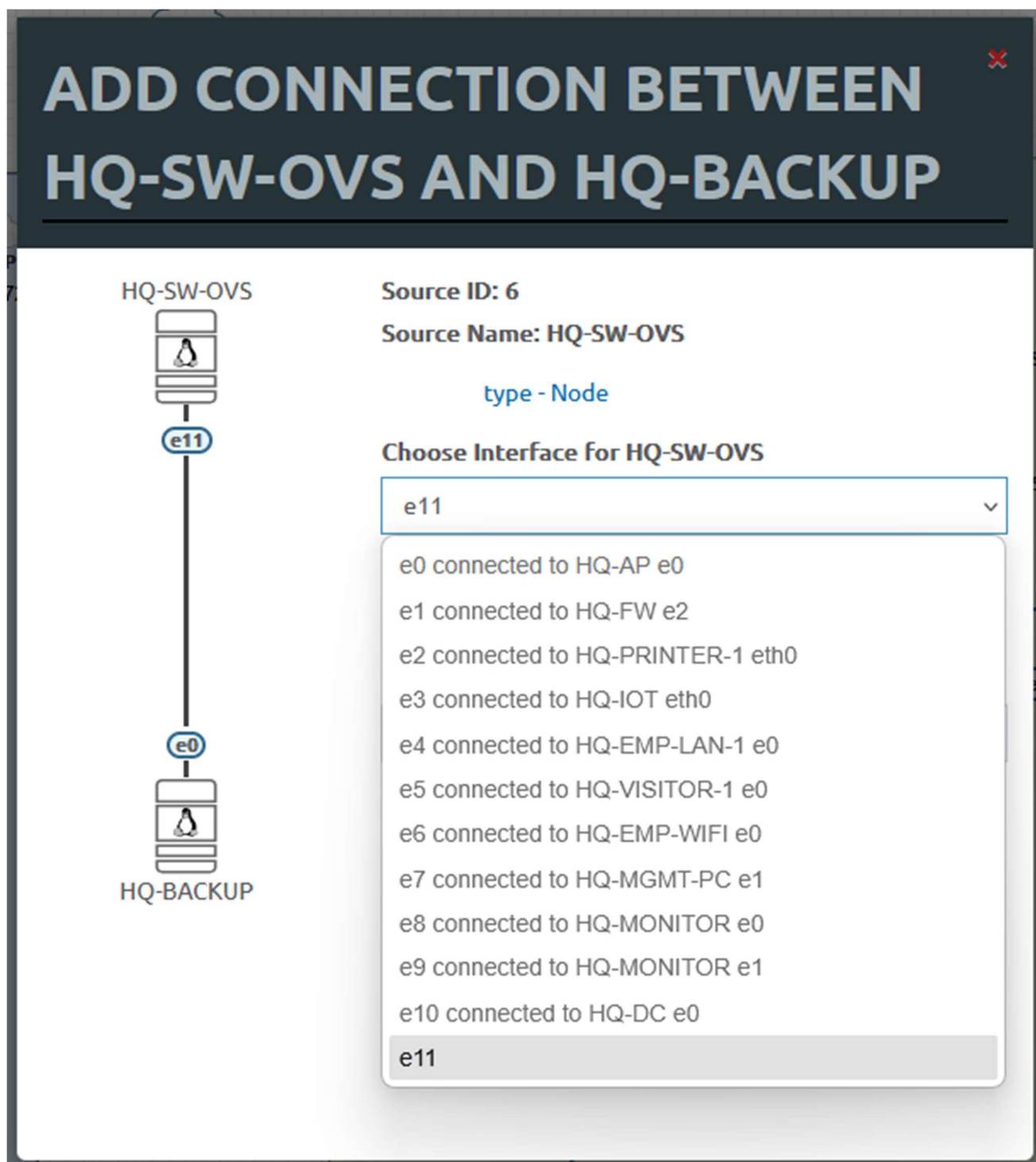


Figure B05 New switch adapter selected

Evidence B06 Existing Open vSwitch configuration

Bridge br0 has an uplink on ens4 already carrying VLAN 91. Existing port assignments were inspected before modification.

```
labadmin@HQ-SW-OVS:~$ sudo ovs-vsctl show
[sudo: authenticate] Password:
fbdb0e19-38fa-482d-9331-c955cd766971
    Bridge br0
        Port ens10
            tag: 99
            Interface ens10
        Port ens7
            tag: 10
            Interface ens7
        Port ens12
            Interface ens12
        Port mgmt99
            tag: 99
            Interface mgmt99
                type: internal
        Port ens5
            tag: 30
            Interface ens5
        Port ens6
            tag: 50
            Interface ens6
        Port ens9
            tag: 11
            Interface ens9
        Port ens4
            trunks: [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
            Interface ens4
        Port ens11
            tag: 90
            Interface ens11
        Port ens8
            tag: 20
            Interface ens8
        Port ens3
            tag: 99
            trunks: [11, 20]
            Interface ens3
        Port br0
            Interface br0
                type: internal
        Port ens13
            tag: 90
            Interface ens13
    ovs_version: "3.7.1"
labadmin@HQ-SW-OVS:~$ _
```

Figure B06 Existing Open vSwitch configuration

Evidence B07 Linux switch interface mapping

ens14 is up and its MAC ends in 0b, consistent with the added e11. EVE labels must not be confused with Linux interface names.

```
labadmin@HQ-SW-OVS:~$ ip -br link
lo                UNKNOWN      00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
ens3              UP          00:50:00:00:06:00 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens4              UP          00:50:00:00:06:01 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens5              UP          00:50:00:00:06:02 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens6              UP          00:50:00:00:06:03 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens7              UP          00:50:00:00:06:04 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens8              UP          00:50:00:00:06:05 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens9              UP          00:50:00:00:06:06 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens10             UP          00:50:00:00:06:07 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens11             UP          00:50:00:00:06:08 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens12             UP          00:50:00:00:06:09 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens13             UP          00:50:00:00:06:0a <BROADCAST,MULTICAST,UP,LOWER_UP>
ens14             UP          00:50:00:00:06:0b <BROADCAST,MULTICAST,UP,LOWER_UP>
ovs-system        DOWN        d6:c1:f3:1c:2c:fd <BROADCAST,MULTICAST>
mgmt99           UNKNOWN      0e:ea:f3:4b:be:82 <BROADCAST,MULTICAST,UP,LOWER_UP>
br0              DOWN        00:50:00:00:06:00 <BROADCAST,MULTICAST>
labadmin@HQ-SW-OVS:~$ _
```

Figure B07 Linux switch interface mapping

Evidence B08 Backup access port configured

ens14 has tag 91 and vlan_mode access. Untagged server traffic is assigned to VLAN 91 by the switch.

```
labadmin@HQ-SW-OVS:~$ ip -br link
lo                UNKNOWN      00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
ens3              UP            00:50:00:00:06:00 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens4              UP            00:50:00:00:06:01 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens5              UP            00:50:00:00:06:02 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens6              UP            00:50:00:00:06:03 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens7              UP            00:50:00:00:06:04 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens8              UP            00:50:00:00:06:05 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens9              UP            00:50:00:00:06:06 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens10             UP            00:50:00:00:06:07 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens11             UP            00:50:00:00:06:08 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens12             UP            00:50:00:00:06:09 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens13             UP            00:50:00:00:06:0a <BROADCAST,MULTICAST,UP,LOWER_UP>
ens14             UP            00:50:00:00:06:0b <BROADCAST,MULTICAST,UP,LOWER_UP>
ovs-system        DOWN        d6:c1:f3:1c:2c:fd <BROADCAST,MULTICAST>
mgmt99           UNKNOWN    0e:ea:f3:4b:be:82 <BROADCAST,MULTICAST,UP,LOWER_UP>
br0              DOWN        00:50:00:00:06:00 <BROADCAST,MULTICAST>
labadmin@HQ-SW-OVS:~$ sudo ovs-vsctl --may-exist add-port br0 ens14 -- set Port ens14 vlan_mode=access tag=91
[sudo: authenticate] Password:
labadmin@HQ-SW-OVS:~$ sudo ovs-vsctl list Port ens14
    _uuid          : 3866d9a7-301b-46c0-a277-83d6292896f6
    bond_active_slave : []
    bond_downdelay   : 0
    bond_fake_iface  : false
    bond_mode        : []
    bond_updelay     : 0
    cvlans           : []
    external_ids     : {}
    fake_bridge      : false
    interfaces       : [c2935839-16cf-4842-a7c4-bc11a3b09e50]
    lacp             : []
    mac              : []
    name             : ens14
    other_config     : {}
    protected        : false
    qos              : []
    rstp_statistics  : {}
    rstp_status      : {}
    statistics       : {}
    status           : {}
    tag              : 91
    trunks           : []
    vlan_mode        : access
labadmin@HQ-SW-OVS:~$
```

Figure B08 Backup access port configured

Evidence B09 Cloned server network settings

ens3 has no IPv4 address or route. Netplan still contains a MAC match from the original image, which must be removed or corrected.

```
labadmin@HQ-BACKUP:~$ ip -br addr
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens3              UP              fe80::250:ff:fe00:2800/64
labadmin@HQ-BACKUP:~$ ip route
labadmin@HQ-BACKUP:~$ ls /etc/netplan
00-installer-config.yaml
labadmin@HQ-BACKUP:~$ sudo cat /etc/netplan/*.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 00:50:00:00:03:00
      set-name: ens3
  version: 2
labadmin@HQ-BACKUP:~$
```

Figure B09 Cloned server network settings

Evidence B10 Configuration backup warning

cp -n warns about portability. It is not a copy failure message; the option avoids overwriting an existing backup.

```
labadmin@HQ-BACKUP:~$ sudo cp -n /etc/netplan/00-installer-config.yaml /etc/netplan/00-installer-config.yaml.bak
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
labadmin@HQ-BACKUP:~$ _
```

Figure B10 Configuration backup warning

Evidence B11 Incorrect initial YAML spacing

version and routes are misaligned and the address lacks a space after the dash. This screenshot records the error, not the final configuration.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
version: 2
  ethernets:
    ens3:
      dhcp4: false
      dhcp6: false
      addresses:
        -192.168.91.10/24
  routes:
    - to: default
      via: 192.168.91.1
```

Figure B11 Incorrect initial YAML spacing

Evidence B12 YAML validation failure

netplan generate rejects inconsistent indentation before applying network changes.

```
labadmin@HQ-BACKUP:~$ sudo chmod 600 /etc/netplan/00-installer-config.yaml
labadmin@HQ-BACKUP:~$ sudo netplan generate
/etc/netplan/00-installer-config.yaml:4:12: Invalid YAML: inconsistent indentation:
  ethernet:
    ^
labadmin@HQ-BACKUP:~$
```

Figure B12 YAML validation failure

Evidence B13 Route key at incorrect level

routes at the document root is rejected. It belongs under ens3, aligned with addresses and nameservers.

```
labadmin@HQ-BACKUP:~$ sudo chmod 600 /etc/netplan/00-installer-config.yaml
labadmin@HQ-BACKUP:~$ sudo netplan generate
/etc/netplan/00-installer-config.yaml:10:1: Error in network definition: unknown key 'routes'
routes:
^
labadmin@HQ-BACKUP:~$
```

Figure B13 Route key at incorrect level

Evidence B14 Static address accepted but ping blocked

192.168.91.10/24 and the default route are present. Four gateway pings receive no replies, requiring a separate connectivity investigation.

```
labadmin@HQ-BACKUP:~$ sudo chmod 600 /etc/netplan/00-installer-config.yaml
labadmin@HQ-BACKUP:~$ sudo netplan generate
labadmin@HQ-BACKUP:~$ sudo netplan try
Do you want to keep these settings?

Press ENTER before the timeout to accept the new configuration

Changes will revert in 116 seconds
Configuration accepted.
labadmin@HQ-BACKUP:~$ ip -br addr
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens3              UP                192.168.91.10/24 fe80::250:ff:fe00:2800/64
labadmin@HQ-BACKUP:~$ ip route
default via 192.168.91.1 dev ens3 proto static
192.168.91.0/24 dev ens3 proto kernel scope link src 192.168.91.10
labadmin@HQ-BACKUP:~$ ping -c 4 192.168.91.1
PING 192.168.91.1 (192.168.91.1) 56(84) bytes of data.

--- 192.168.91.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3110ms

labadmin@HQ-BACKUP:~$ _
```

Figure B14 Static address accepted but ping blocked

Evidence B15 Gateway neighbour resolved

A MAC is cached for 192.168.91.1. STALE is an ordinary neighbour-cache state and supports successful earlier ARP exchange.

```
labadmin@HQ-BACKUP:~$ ip neigh show 192.168.91.1
192.168.91.1 dev ens3 lladdr 50:00:00:01:00:02 STALE
labadmin@HQ-BACKUP:~$ _
```

Figure B15 Gateway neighbour resolved

Evidence B16 Gateway ICMP rule

The rule allows an IPv4 echo request from HQ-BACKUP to its gateway and enables logging. Single addresses use Address or Alias.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

BACKUP_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

ICMP

Choose which IP protocol this rule should match.

ICMP Subtypes

Echo request

Information reply

Information request

IPv6 I am here

For ICMP rules on IPv4, one or more of these ICMP subtypes may be specified.

Source

Source

☐ Invert match

Address or Alias

192.168.91.10

/

Destination

Destination

☐ Invert match

Address or Alias

192.168.91.1

/

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Allow HQ-BACKUP gateway ping

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Save

Figure B16 Gateway ICMP rule

Evidence B17 Gateway ping succeeds

Four replies and zero packet loss follow the firewall allowance. This verifies the tested gateway path, not every routed service.

```
labadmin@HQ-BACKUP:~$ ip neigh show 192.168.91.1
192.168.91.1 dev ens3 lladdr 50:00:00:01:00:02 STALE
labadmin@HQ-BACKUP:~$ ping -c 4 192.168.91.1
PING 192.168.91.1 (192.168.91.1) 56(84) bytes of data.
64 bytes from 192.168.91.1: icmp_seq=1 ttl=64 time=1.78 ms
64 bytes from 192.168.91.1: icmp_seq=2 ttl=64 time=1.13 ms
64 bytes from 192.168.91.1: icmp_seq=3 ttl=64 time=1.13 ms
64 bytes from 192.168.91.1: icmp_seq=4 ttl=64 time=1.33 ms

--- 192.168.91.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.125/1.338/1.776/0.265 ms
labadmin@HQ-BACKUP:~$ _
```

Figure B17 Gateway ping succeeds

Evidence B18 Operating system and initial storage

Ubuntu 26.04 LTS is shown; only a 30 GB OS disk exists and no DNS server is configured.

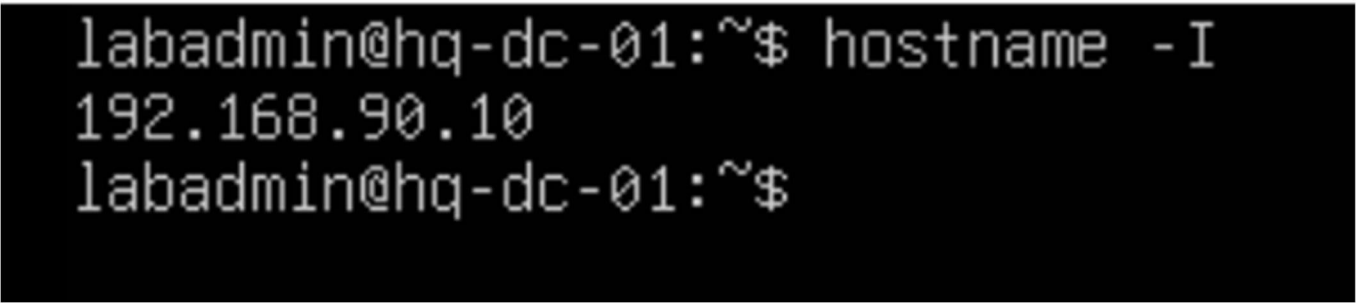
```
labadmin@HQ-BACKUP:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens3)
    Current Scopes: none
    Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Default Route: no
labadmin@HQ-BACKUP:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME                                SIZE FSTYPE MOUNTPOINTS
fd0                                 4K
sr0                                1024M
vda                                 30G
├─vda1                              1M
├─vda2                              2G ext4      /boot
├─vda3                              28G LVM2_member
└─ubuntu--vg-ubuntu--lv            14G ext4      /
labadmin@HQ-BACKUP:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
labadmin@HQ-BACKUP:~$
```

Figure B18 Operating system and initial storage

Evidence B19 HQ DC address

hostname -I identifies HQ-DC at 192.168.90.10 for subsequent DNS and backup access.



```
labadmin@hq-dc-01:~$ hostname -I
192.168.90.10
labadmin@hq-dc-01:~$
```

Figure B19 HQ DC address

Evidence B20 DNS listeners on HQ DC

Port 53 listens on TCP and UDP. This supports using HQ-DC as DNS, but reachability still needs a client test.

```
labadmin@hq-dc-01:~$ hostname -I
192.168.90.10
labadmin@hq-dc-01:~$ sudo ss -ltnp 'sport = :53'
[sudo: authenticate] Password:
Netid   State   Recv-Q   Send-Q   Local Address:Port   Peer Address:Port   Process
udp     UNCONN  0         0         0.0.0.0:53           0.0.0.0:*           users:("dns[master]",pid=23590,fd=51))
udp     UNCONN  0         0         [::]:53             [::]:*              users:("dns[master]",pid=23590,fd=49))
tcp     LISTEN  0         10        0.0.0.0:53           0.0.0.0:*           users:("dns[master]",pid=23590,fd=50))
tcp     LISTEN  0         10        [::]:53             [::]:*              users:("dns[master]",pid=23590,fd=47))
labadmin@hq-dc-01:~$ _
```

Figure B20 DNS listeners on HQ DC

Evidence B21 DNS rule before port correction

The destination range starts at any and ends at 53. It was corrected to 53 through 53 before use.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

BACKUP_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP/UDP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Network

192.168.91.10

/

32

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

Network

192.168.90.10

/

32

Destination Port Range

any

From

Custom

DNS (53)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Allow HQ-BACKUP DNS to HQ-DC

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Save

Figure B21 DNS rule before port correction

Evidence B22 Corrected DNS rule

TCP and UDP destination ports are both 53; source and destination are individual host addresses.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

BACKUP_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP/UDP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Address or Alias

192.168.91.10

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

Address or Alias

192.168.90.10

/

Destination Port Range

DNS (53)

From

Custom

To

DNS (53)

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Allow HQ-BACKUP DNS to HQ-DC

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

1790654805

Figure B22 Corrected DNS rule

Evidence B23 DNS client configuration still absent

The resolver reports no appropriate name servers. A firewall rule alone cannot configure the client resolver.

```
labadmin@HQ-BACKUP:~$ resolvectl status ens3
Link 2 (ens3)
  Current Scopes: none
  Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no
labadmin@HQ-BACKUP:~$ resolvectl query ubuntu.com
ubuntu.com: resolve call failed: No appropriate name servers or networks for name found
labadmin@HQ-BACKUP:~$ _
```

Figure B23 DNS client configuration still absent

Evidence B24 Final Netplan configuration

The nameservers block adds 192.168.90.10 alongside the static IPv4 address and gateway.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  version: 2
  ethernets:
    ens3:
      dhcp4: false
      dhcp6: false
      addresses:
        - 192.168.91.10/24
      nameservers:
        addresses:
          - 192.168.90.10
      routes:
        - to: default
          via: 192.168.91.1
```

Figure B24 Final Netplan configuration

Evidence B25 Successful DNS query

The configuration is accepted and ubuntu.com resolves through ens3 with DNS server 192.168.90.10.

```
labadmin@HQ-BACKUP:~$ sudo netplan generate
labadmin@HQ-BACKUP:~$ sudo netplan try
Do you want to keep these settings?

Press ENTER before the timeout to accept the new configuration

Changes will revert in 119 seconds
Configuration accepted.
labadmin@HQ-BACKUP:~$ resolvectl status ens3
Link 2 (ens3)
  Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    DNS Servers: 192.168.90.10
    Default Route: yes
labadmin@HQ-BACKUP:~$ resolvectl query ubuntu.com
ubuntu.com: 185.125.190.29      -- link: ens3
             185.125.190.20      -- link: ens3
             185.125.190.21      -- link: ens3
             2620:2d:4000:1::28    -- link: ens3
             2620:2d:4000:1::26    -- link: ens3
             2620:2d:4000:1::27    -- link: ens3

-- Information acquired via protocol DNS in 30.2ms.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: network
labadmin@HQ-BACKUP:~$ _
```

Figure B25 Successful DNS query

Evidence B26 Initial package update failure

IPv6 has no route; IPv4 HTTP connections also time out. Existing upgrade counts do not establish a successful update.

```
Err:1 http://au.archive.ubuntu.com/ubuntu/resolute InRelease
  Cannot initiate the connection to au.archive.ubuntu.com:80 (2001:388:30bc:cafe::beef). - connect (101: Network is unreachable)
  Could not connect to au.archive.ubuntu.com:80 (202.158.214.106), connection timed out
  Cannot initiate the connection to au.archive.ubuntu.com:80 (2001:388:30bc:cafe::beef). - connect (101: Network is unreachable)
Err:4 http://security.ubuntu.com/ubuntu/resolute-security InRelease
  Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (101: Network is unreachable) Cannot
  initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (101: Network is unreachable) Cannot initia
  te the connection to security.ubuntu.com:80 (2620:2d:4000:1::103). - connect (101: Network is unreachable) Cannot initiate the c
  onnection to security.ubuntu.com:80 (2620:2d:4000:1::102). - connect (101: Network is unreachable) Cannot initiate the connectio
  n to security.ubuntu.com:80 (2620:2d:4000:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to sec
  urity.ubuntu.com:80 (2a06:bc80:0:1000::17). - connect (101: Network is unreachable) Cannot initiate the connection to security.u
  buntu.com:80 (2620:2d:4002:1::102). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.co
  m:80 (2620:2d:4002:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a
  06:bc80:0:1000::18). - connect (101: Network is unreachable) Could not connect to security.ubuntu.com:80 (91.189.91.82), connect
  ion timed out Could not connect to security.ubuntu.com:80 (91.189.92.22), connection timed out Could not connect to security.ubu
  ntubuntu.com:80 (185.125.190.82), connection timed out Could not connect to security.ubuntu.com:80 (185.125.190.83), connection timed
  out Could not connect to security.ubuntu.com:80 (91.189.91.83), connection timed out Could not connect to security.ubuntu.com:8
  0 (91.189.92.24), connection timed out Could not connect to security.ubuntu.com:80 (91.189.92.23), connection timed out Could no
  t connect to security.ubuntu.com:80 (185.125.190.81), connection timed out Could not connect to security.ubuntu.com:80 (91.189.9
  1.81), connection timed out
  Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (101: Network is unreachable) Cannot
  initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (101: Network is unreachable) Cannot initia
  te the connection to security.ubuntu.com:80 (2620:2d:4000:1::103). - connect (101: Network is unreachable) Cannot initiate the c
  onnection to security.ubuntu.com:80 (2620:2d:4000:1::102). - connect (101: Network is unreachable) Cannot initiate the connectio
  n to security.ubuntu.com:80 (2620:2d:4000:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to sec
  urity.ubuntu.com:80 (2a06:bc80:0:1000::17). - connect (101: Network is unreachable) Cannot initiate the connection to security.u
  buntu.com:80 (2620:2d:4002:1::102). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.co
  m:80 (2620:2d:4002:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a
  06:bc80:0:1000::18). - connect (101: Network is unreachable)
54 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://au.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Cannot initiate the connection to au.arch
ive.ubuntu.com:80 (2001:388:30bc:cafe::beef). - connect (101: Network is unreachable)
Warning: Failed to fetch http://au.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Cannot initiate the connection to
au.archive.ubuntu.com:80 (2001:388:30bc:cafe::beef). - connect (101: Network is unreachable)
Warning: Failed to fetch http://au.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Cannot initiate the connection
to au.archive.ubuntu.com:80 (2001:388:30bc:cafe::beef). - connect (101: Network is unreachable)
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Cannot initiate the connection to
security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (101: Network is unreachable) Cannot initiate the connection to security
.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu
.com:80 (2620:2d:4000:1::103). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80
(2620:2d:4000:1::102). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d
:4000:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:100
0::17). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::102).
- connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::101). - connec
t (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::18). - connect (101:
Network is unreachable)
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
labadmin@HQ-BACKUP:~$ _
```

Figure B26 Initial package update failure

Evidence B27 Temporary HTTP allowance

Only HQ-BACKUP is allowed IPv4 TCP destination port 80 to any address. This temporary rule is broader than a repository-only policy.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

BACKUP_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Address or Alias

192.168.91.10

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

Any

Destination Address

Destination Port Range

HTTP (80)

From

Custom

To

HTTP (80)

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Temporary HQ-BACKUP package downloads

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Save

Figure B27 Temporary HTTP allowance

Evidence B28 Package indexes downloaded

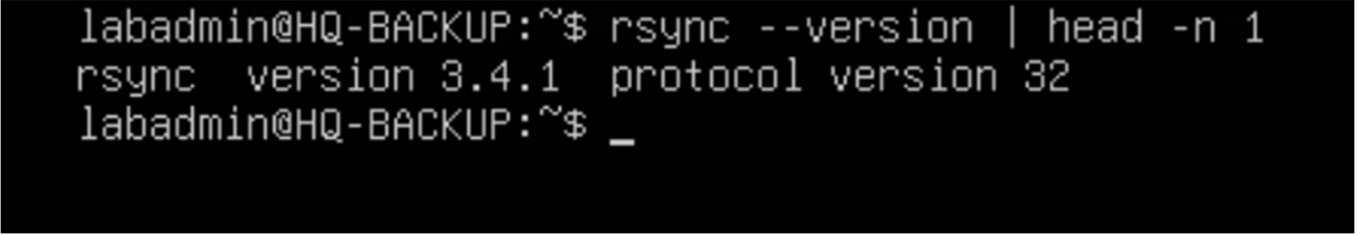
apt update with ForceIPv4 succeeds after the allowance. It refreshes indexes; the displayed 173 upgrades are not installed by this command.

```
labadmin@HQ-BACKUP:~$ sudo apt -o Acquire::ForceIPv4=true update
Hit:1 http://au.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://au.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://au.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://au.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [679 kB]
Get:5 http://au.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [162 kB]
Get:6 http://au.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:7 http://au.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [457 kB]
Get:8 http://au.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [90.8 kB]
Get:9 http://au.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [308 kB]
Get:10 http://au.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [98.9 kB]
Get:11 http://au.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [190 kB]
Get:12 http://au.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:13 http://au.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3,172 B]
Get:14 http://au.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3,132 B]
Get:15 http://au.archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7,988 B]
Get:16 http://au.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Get:17 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [543 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [127 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [443 kB]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [87.8 kB]
Get:23 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [187 kB]
Get:24 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [59.3 kB]
Get:25 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:26 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [10.8 kB]
Get:27 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2,844 B]
Fetched 4,072 kB in 4s (1,112 kB/s)
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
labadmin@HQ-BACKUP:~$ _
```

Figure B28 Package indexes downloaded

Evidence B29 Rsync version confirmed

rsync 3.4.1 is installed. The later demonstrated archive transfer uses SCP, not rsync.

A terminal window with a black background and white text. The prompt is 'labadmin@HQ-BACKUP:~\$'. The command entered is 'rsync --version | head -n 1'. The output is 'rsync version 3.4.1 protocol version 32'. The prompt is repeated as 'labadmin@HQ-BACKUP:~\$ _' with an underscore cursor.

```
labadmin@HQ-BACKUP:~$ rsync --version | head -n 1
rsync version 3.4.1 protocol version 32
labadmin@HQ-BACKUP:~$ _
```

Figure B29 Rsync version confirmed

Evidence B30 HQ BACKUP node settings

Node 40 uses the Linux image, two vCPUs, 4096 MB RAM and one interface. The editor has no direct disk-count control.

EDIT NODE

Template

Linux

ID

40

Image

linux-ubuntu-server-26.04

Name/prefix

HQ-BACKUP

Icon

Server-2D-Linux-S.svg

UUID

fddc5e07-9a43-46c2-b23f-13bb0e6e001e

CPU Limit

CPU

2

RAM (MB)

4096

Ethernets

1

First Eth MAC Address

00:50:00:00:28:00

QEMU Version

2.12.0

QEMU Arch

x86_64

QEMU Nic

virtio-net-pci

QEMU custom options

-machine type=pc,accel=kvm -vga std -usbdevice tablet -boot order=cd -cpu host

Startup configuration

None

Delay (s)

0

Console

vnc

Left

828

Top

516

Save

Cancel

Figure B30 HQ BACKUP node settings

Evidence B31 EVE storage and node directory

The host reports 295 GB available and locates node 40 under the lab working directory. Capacity is a point-in-time observation.

```
root@prime-core:~# df -h /opt/unetlab
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 490G  176G   295G  38% /
root@prime-core:~# sudo find /opt/unetlab/tmp -type d -name 40
/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40
root@prime-core:~#
```

Figure B31 EVE storage and node directory

Evidence B32 System disk and stop log

virtioa.qcow2 is the existing OS disk. The wrapper log records a stopped child; it is not evidence of the later disk attachment itself.

```
root@prime-core:~# cd /opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# ls -lh
total 566M
drwxr-sr-x  4 root root 4.0K Aug 13 08:55 dev
drwxr-sr-x  2 root root 4.0K May  9 2022 jail
lrwxrwxrwx  1 root root   8 May  3 2022 lib -> jail/lib
lrwxrwxrwx  1 root root  10 May  3 2022 lib64 -> jail/lib64
drwxr-sr-x 17 root root 4.0K Aug 15 06:26 opt
drwxr-sr-x  4 root root 4.0K Aug 13 08:55 sys
lrwxrwxrwx  1 root root   8 May  3 2022 usr -> jail/usr
-rw-r--r--  1 root unl 566M Sep 29 04:23 virtioa.qcow2
-rw-rw-r--  1 root unl  171 Sep 29 04:23 wrapper.txt
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# cat wrapper.txt
29/8 3:13:21.246 INF   Tennant_id = 0
29/8 3:13:21.246 INF   Device_id = 40
29/8 4:23:42.540 INF   Caught SIGTERM, killing child.
29/8 4:23:42.541 ERR   Child is no more running.
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# _
```

Figure B32 System disk and stop log

Evidence B33 Disk creation syntax failure

The command omitted the output filename. qemu-img interpreted 100G as a filename and reported that a size was missing.

```
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# if [ ! -e virtiob.qcow2 ]; then
> /opt/qemu/bin/qemu-img create -f qcow2 100G
> fi
qemu-img: 100G: Image creation needs a size parameter
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40#
```

Figure B33 Disk creation syntax failure

Evidence B34 Additional disk created

The guarded command creates virtiob.qcow2 with a 100 GB virtual capacity and applies root:unl ownership and mode 644.

```
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# if [ ! -e virtiob.qcow2 ]; then
> /opt/qemu/bin/qemu-img create -f qcow2 virtiob.qcow2 100G
> fi
Formatting 'virtiob.qcow2', fmt=qcow2 size=107374182400 encryption=off cluster_size=65536 lazy_refcounts=off refcount_bits=16
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# chown root:unl virtiob.qcow2
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40# chmod 644 virtiob.qcow2
root@prime-core:/opt/unetlab/tmp/0/8183fac7-5405-46a8-9b15-9619a5c56489/40#
```

Figure B34 Additional disk created

Evidence B35 Disk file not yet attached

The guest still sees only vda. Creating a file in the node folder did not automatically attach it in this setup.

```
labadmin@HQ-BACKUP:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME                                SIZE FSTYPE  MOUNTPOINTS
fd0                                  4K
sr0                                1024M
vda                                 30G
├─vda1                               1M
├─vda2                               2G ext4      /boot
└─vda3                               28G LVM2_member
    └─ubuntu--vg-ubuntu--lv         14G ext4      /
labadmin@HQ-BACKUP:~$ _
```

Figure B35 Disk file not yet attached

Evidence B36 Second disk detected

After the explicit QEMU drive option, the guest sees the new blank 100 GB vdb. This distinguishes it from the 30 GB OS disk.

```
labadmin@HQ-BACKUP:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME                                SIZE FSTYPE MOUNTPOINTS
fd0                                  4K
sr0                                 1024M
vda                                  30G
├─vda1                               1M
├─vda2                               2G ext4      /boot
├─vda3                               28G LVM2_member
│   └─ubuntu--vg-ubuntu--lv         14G ext4      /
vdb                                 100G
labadmin@HQ-BACKUP:~$ _
```

Figure B36 Second disk detected

Evidence B37 Filesystem creation and mount

ext4 is created only on vdb, labelled PRIMECORE_BACKUP, mounted at /srv/backups and identified by UUID.

```
labadmin@HQ-BACKUP:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME                                SIZE FSTYPE MOUNTPOINTS
fd0                                  4K
sr0                                  1024M
vda                                  30G
├─vda1                               1M
├─vda2                               2G ext4      /boot
├─vda3                               28G LVM2_member
└─ubuntu--vg-ubuntu--lv             14G ext4      /
vdb                                  100G
labadmin@HQ-BACKUP:~$ sudo mkfs.ext4 -L PRIMECORE_BACKUP /dev/vdb
[sudo: authenticate] Password:
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 26214400 4k blocks and 6553600 inodes
Filesystem UUID: 4ee001f1-8953-40b4-a3cf-20173e67a8ed
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000, 7962624, 11239424, 20480000, 23887872

Allocating group tables: done
Writing inode tables: done
Creating journal (131072 blocks): done
Writing superblocks and filesystem accounting information: done

labadmin@HQ-BACKUP:~$ sudo mkdir -p /srv/backups
labadmin@HQ-BACKUP:~$ sudo mount /dev/vdb /srv/backups
labadmin@HQ-BACKUP:~$ sudo blkid /dev/vdb
/dev/vdb: LABEL="PRIMECORE_BACKUP" UUID="4ee001f1-8953-40b4-a3cf-20173e67a8ed" BLOCK_SIZE="4096" TYPE="ext4"
labadmin@HQ-BACKUP:~$
```

Figure B37 Filesystem creation and mount

Evidence B38 Persistent mount entry

The UUID entry targets /srv/backups with ext4 defaults and filesystem-check pass 2. Other existing fstab entries remain intact.

```
GNU nano 8.7.1 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/ubuntu-vg/ubuntu-lv during curtin installation
/dev/disk/by-id/dm-uuid-LVM-asLWMMCIx26dfWJWx50g1XtqK9qX1i0e2RGBmzF9I1YK8csaW6vjnCHTxJRsh3IA / ext4 defaults 0 1
# /boot was on /dev/vda2 during curtin installation
/dev/disk/by-uuid/d49d8799-0502-41c5-a3a0-902c2f123d2d /boot ext4 defaults 0 1
/swap.img none swap sw 0 0
UUID=4ee001f1-8953-40b4-a3cf-20173e67a8ed /srv/backups ext4 defaults 0 2_
```

Figure B38 Persistent mount entry

Evidence B39 Mount validation and Restic version

findmnt reports zero errors; its warning is for the existing swap file. vdb is mounted with about 93 GB available; Restic is version 0.18.1.

```
labadmin@HQ-BACKUP:~$ sudo systemctl daemon-reload
labadmin@HQ-BACKUP:~$ sudo findmnt --verify
none
[W] non-bind mount source /swap.img is a directory or regular file

0 parse errors, 0 errors, 1 warning
labadmin@HQ-BACKUP:~$ df -h /srv/backups
Filesystem      Size  Used Avail Use% Mounted on
/dev/vdb         98G   2.1M   93G   1% /srv/backups
labadmin@HQ-BACKUP:~$ restic version
restic 0.18.1 compiled with go1.25.0 on linux/amd64
labadmin@HQ-BACKUP:~$
```

Figure B39 Mount validation and Restic version

Evidence B40 Repository initialisation and check

The encrypted repository is created and its initial check reports no errors. It contains zero snapshots at this stage.

```
labadmin@HQ-BACKUP:~$ sudo chmod 700 /srv/backups
labadmin@HQ-BACKUP:~$ sudo mountpoint -q /srv/backups && sudo restic -r /srv/backups/primecore init
enter password for new repository:
enter password again:
created restic repository 4104af51e8 at /srv/backups/primecore

Please note that knowledge of your password is required to access
the repository. Losing your password means that your data is
irrecoverably lost.
labadmin@HQ-BACKUP:~$ sudo restic -r /srv/backups/primecore check
using temporary cache in /tmp/restic-check-cache-672630677
create exclusive lock for repository
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
created new cache in /tmp/restic-check-cache-672630677
load indexes
[0:00]      0 index files loaded
check all packs
check snapshots, trees and blobs
[0:00]      0 snapshots
no errors were found
labadmin@HQ-BACKUP:~$ _
```

Figure B40 Repository initialisation and check

Evidence B41 Local file recovery test

Snapshot 157bfc4c restores successfully. diff reports identical contents and stat confirms labadmin:labadmin ownership and mode 640.

```
labadmin@HQ-BACKUP:~$ mkdir -p ~/primecore-backup-test
labadmin@HQ-BACKUP:~$ printf 'PrimeCore backup and restore test\n' > ~/primecore-backup-test/test.txt
labadmin@HQ-BACKUP:~$ chmod 640 ~/primecore-backup-test/test.txt
labadmin@HQ-BACKUP:~$ sudo mountpoint -q /srv/backups && sudo restic -r /srv/backups/primecore backup /home/labadmin/primecore-backup-test --tag restore-test
[sudo: authenticate] Password:
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
created new cache in /root/.cache/restic
no parent snapshot found, will read all files
[0:00] 0 index files loaded

Files:          1 new,      0 changed,      0 unmodified
Dirs:           3 new,      0 changed,      0 unmodified
Added to the repository: 1.422 KiB (1.271 KiB stored)

processed 1 files, 34 B in 0:00
snapshot 157bfc4c saved
labadmin@HQ-BACKUP:~$ sudo restic -r /srv/backups/primecore restore latest --tag restore-test --target /var/tmp/primecore-restore-test
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
[0:00] 100.00% 1 / 1 index files loaded
restoring snapshot 157bfc4c of [/home/labadmin/primecore-backup-test] at 2026-09-29 22:58:00.591658178 +0000 UTC by root@HQ-BACKUP to /var/tmp/primecore-restore-test
Summary: Restored 4 files/dirs (34 B) in 0:00
labadmin@HQ-BACKUP:~$ sudo diff -s /home/labadmin/primecore-backup-test/test.txt /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt
Files /home/labadmin/primecore-backup-test/test.txt and /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt are identical
labadmin@HQ-BACKUP:~$ sudo stat -c '%U:%G %a %n' /home/labadmin/primecore-backup-test/test.txt /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt
labadmin:labadmin 640 /home/labadmin/primecore-backup-test/test.txt
labadmin:labadmin 640 /var/tmp/primecore-restore-test/home/labadmin/primecore-backup-test/test.txt
labadmin@HQ-BACKUP:~$
```

Figure B41 Local file recovery test

Evidence B42 Samba version and server role

Samba 4.23.6 and the Active Directory domain controller role are confirmed. This determines the supported backup method.

```
labadmin@hq-dc-01:~$ sudo samba-tool --version
4.23.6-Ubuntu-4.23.6+dfsg-1ubuntu2.2
labadmin@hq-dc-01:~$ sudo testparm -s --parameter-name='server role'
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

active directory domain controller
labadmin@hq-dc-01:~$ _
```

Figure B42 Samba version and server role

Evidence B43 Supported Samba backup options

The online backup help lists server, target directory and credentials options. Secret values and SYSVOL are not excluded from this backup.

```
Options:
-h, --help                show this help message and exit
--server=SERVER           The DC to backup
--targetdir=TARGETDIR     Directory to write the backup file to
--no-secrets              Exclude secret values from the backup created
--backend-store=BACKENDSTORE
                          Specify the database backend to be used (default is
                          tdb)
--no-sysvol               Exclude sysvol from the backup created
--color=always|never|auto
                          use colour if available (default: auto)

Version Options:
-V, --version             Display version number
                          (4.23.6-Ubuntu-4.23.6+dfsg-1ubuntu2.2)

Credentials Options:
--simple-bind-dn=DN        DN to use for a simple bind
--password=PASSWORD       Password
-U USERNAME, --username=USERNAME
                          Username
-W WORKGROUP, --workgroup=WORKGROUP
                          Workgroup
-N, --no-pass             Don't ask for a password
--ipaddress=IPADDRESS     IP address of server
-P, --machine-pass        Use stored machine account password
--use-kerberos=desired|required|off
                          Use Kerberos authentication
--use-krb5-ccache=KRB5CCNAME
                          Kerberos Credentials cache
-A AUTHFILE, --authentication-file=AUTHFILE
                          Authentication file
-k KERBEROS, --kerberos=KERBEROS
                          DEPRECATED: Migrate to --use-kerberos

Samba Common Options:
-s FILE, --configfile=FILE
                          Configuration file
-d DEBUGLEVEL, --debuglevel=DEBUGLEVEL
                          debug level
--option=OPTION           set smb.conf option from command line
--realm=REALM             set the realm name
labadmin@hq-dc-01:~$ _
```

Figure B43 Supported Samba backup options

Evidence B44 SSH listener on HQ DC

TCP port 22 listens on IPv4 and IPv6 via systemd. Subsequent SCP success confirms usable SSH access.

```
labadmin@hq-dc-01:~$ sudo ss -ltnp 'sport = :22'
```

State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*	users:(("systemd",pid=1,fd=229))
LISTEN	0	4096	:::22	:::*	users:(("systemd",pid=1,fd=230))

```
labadmin@hq-dc-01:~$
```

Figure B44 SSH listener on HQ DC

Evidence B45 Samba online backup completes

Directory partitions and DNS zones are replicated, SYSVOL is backed up and a primecore.test archive is created. Temporary Kerberos guidance is not an instruction to change the live DC.

```
INFO 2026-09-29 23:19:46,295 pid:93707 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2425: A Kerberos configuration suitable for Samba AD has been generated at /var/backups/primecore-ad/tmp722jy3on/private/krb5.conf
INFO 2026-09-29 23:19:46,295 pid:93707 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2427: Merge the contents of this file with your system krb5.conf or replace it with this one. Do not create a symlink!
Provision OK for domain DN DC=primecore,DC=test
INFO 2026-09-29 23:19:46,311 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #973: Starting replication
Using DS_BIND_GUID_M2K3
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[402/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[804/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1206/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1608/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1770/1770] linked_values[0/0]
Analyze and apply schema objects
Partition [CN=Configuration,DC=primecore,DC=test] objects[402/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[804/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1206/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1608/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1731/1731] linked_values[54/54]
Replicating critical objects from the base DN of the domain
Partition [DC=primecore,DC=test] objects[97/97] linked_values[23/23]
Partition [DC=primecore,DC=test] objects[288/288] linked_values[24/24]
Done with always replicated NC (base, config, schema)
Replicating DC=DomainDnsZones,DC=primecore,DC=test
Partition [DC=DomainDnsZones,DC=primecore,DC=test] objects[40/40] linked_values[0/0]
Replicating DC=ForestDnsZones,DC=primecore,DC=test
Partition [DC=ForestDnsZones,DC=primecore,DC=test] objects[18/18] linked_values[0/0]
INFO 2026-09-29 23:19:50,122 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1094: Committing SAM database - this may take some time
Repacking database from v1 to v2 format (first record CN=Print-Start-Time,CN=Schema,CN=Configuration,DC=primecore,DC=test)
Repack: re-packed 10000 records so far
Repacking database from v1 to v2 format (first record CN=IntelliMirrorGroup-Display,CN=C0A,CN=DisplaySpecifiers,CN=Configuration,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record DC=l.root-servers.net,DC=RootDNSServers,CN=MicrosoftDNS,DC=DomainDnsZones,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record DC=_ldap._tcp.dc,DC=_msdcs.primecore.test,CN=MicrosoftDNS,DC=ForestDnsZones,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record CN=Dfs-Configuration,CN=System,DC=primecore,DC=test)
INFO 2026-09-29 23:19:52,348 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1114: Committed SAM database
INFO 2026-09-29 23:19:52,354 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1692: Setting IsSynchronized and dsServiceName
INFO 2026-09-29 23:19:52,379 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1656: Cloned domain PRIMECORE (SID S-1-5-21-369086394-685125275-528371242)
INFO 2026-09-29 23:19:52,441 pid:93707 /usr/lib/python3/dist-packages/samba/netcmd/domain/backup.py #290: Backing up sysvol files (via SMB)...
INFO 2026-09-29 23:19:53,021 pid:93707 /usr/lib/python3/dist-packages/samba/netcmd/domain/backup.py #147: Creating backup file /var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2...
labadmin@hq-dc-01:~$
```

Figure B45 Samba online backup completes

Evidence B46 AD backup archive listing

The archive is approximately 1.4 MB and initially has mode 644 inside the restricted staging directory. File permissions were subsequently tightened.

```
labadmin@hq-dc-01:~$ sudo ls -lh /var/backups/primecore-ad
total 1.4M
-rw-r--r-- 1 root root 1.4M Sep 29 23:19 samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2
labadmin@hq-dc-01:~$ _
```

Figure B46 AD backup archive listing

Evidence B47 Archive check dependency missing

chmod restricts the archive. tar -tjf fails because the external bzip2 program is absent, not because corruption was established.

```
labadmin@hq-dc-01:~$ sudo ls -lh /var/backups/primecore-ad
total 1.4M
-rw-r--r-- 1 root root 1.4M Sep 29 23:19 samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2
labadmin@hq-dc-01:~$ sudo find /var/backups/primecore-ad -maxdepth 1 -type f -name '*.tar.bz2' -exec chmod 600 {} +
[sudo: authenticate] Password:
labadmin@hq-dc-01:~$ sudo tar -tjf /var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2 > /dev/null && echo "Archive readable"
tar (child): bzip2: Cannot exec: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
labadmin@hq-dc-01:~$
labadmin@hq-dc-01:~$ _
```

Figure B47 Archive check dependency missing

Evidence B48 Python archive validation succeeds

Python reads the full compressed stream and the tar catalogue. The result is a passed compressed-data check and 30 entries.

```
labadmin@hq-dc-01:~$ sudo python3 - <<'PY'
import bz2
import tarfile
from pathlib import Path

p = Path("/var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2")

with bz2.open(p, "rb") as archive:
    while archive.read(1024 * 1024):
        pass
> print("Compressed data check passed")
>
> with tarfile.open(p, "r:bz2") as archive:
>     print(f"Archive readable: {len(archive.getmembers())} entries")
> PY
[sudo: authenticate] Password:
Compressed data check passed
Archive readable: 30 entries
labadmin@hq-dc-01:~$
```

Figure B48 Python archive validation succeeds

Evidence B49 Restricted export and source identity

The archive is exported for labadmin with mode 600 in a mode 700 directory. SHA256 and the SSH host fingerprint provide transfer checks.

```
labadmin@hq-dc-01:~$ sudo install -d -o labadmin -m 700 /home/labadmin/backup-export
[sudo: authenticate] Password:
labadmin@hq-dc-01:~$ sudo install -o labadmin -m 600 /var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.9
05930.tar.bz2 /home/labadmin/backup-export/hq-dc-ad.tar.bz2
labadmin@hq-dc-01:~$ sha256sum /home/labadmin/backup-export/hq-dc-ad.tar.bz2
b9c6e8116f1b7fff0a2e5aa373244d70c5d58744574d9a2ec3ac6e51a32234f /home/labadmin/backup-export/hq-dc-ad.tar.bz2
labadmin@hq-dc-01:~$ sudo ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:fcKf174/PuWsfB248mS7pb0/H10nmpTADWU8P/SgTY4 root@srv-lnx-01 (ED25519)
labadmin@hq-dc-01:~$ _
```

Figure B49 Restricted export and source identity

Evidence B50 Secure transfer and matching checksum

An accidentally combined command is corrected. SCP completes, the host fingerprint matches and the destination SHA256 matches the source.

```
labadmin@HQ-BACKUP:~$ mkdir -p ~/backup-staging/hq-dc
labadmin@HQ-BACKUP:~$ chmod 700 ~/backup-staging ~/backup-staging/hq-dc scp -o HostKeyAlgorithms=ssh-ed25519 labadmin@192.168.90
.10:/home/labadmin/backup-export/hq-dc-ad.tar.bz2 ~/backup-staging/hq-dc/
chmod: cannot access '700': No such file or directory
chmod: cannot access '/home/labadmin/backup-staging/hq-dc': No such file or directory
chmod: cannot access 'scp': No such file or directory
chmod: cannot access 'HostKeyAlgorithms=ssh-ed25519': No such file or directory
chmod: cannot access 'labadmin@192.168.90.10:/home/labadmin/backup-export/hq-dc-ad.tar.bz2': No such file or directory
chmod: cannot access '/home/labadmin/backup-staging/hq-dc/': No such file or directory
labadmin@HQ-BACKUP:~$ chmod 700 ~/backup-staging ~/backup-staging/hq-dc
labadmin@HQ-BACKUP:~$ scp -o HostKeyAlgorithms=ssh-ed25519 labadmin@192.168.90.10:/home/labadmin/backup-export/hq-dc-ad.tar.bz2
~/backup-staging/hq-dc/
The authenticity of host '192.168.90.10 (192.168.90.10)' can't be established.
ED25519 key fingerprint is: SHA256:fcWf174/PuWsfB248mS7pb0/H10nmptA0WU8P/SgTY4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.90.10' (ED25519) to the list of known hosts.
labadmin@192.168.90.10's password:
hq-dc-ad.tar.bz2                                     100% 1428KB  13.1MB/s   00:00
labadmin@HQ-BACKUP:~$ sha256sum ~/backup-staging/hq-dc/hq-dc-ad.tar.bz2
b9c6e8116f1b7fff0a2e5aa3732444d70c5d58744574d9a2ec3ac6e51a32234f  /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2
labadmin@HQ-BACKUP:~$ _
```

Figure B50 Secure transfer and matching checksum

Evidence B51 AD archive stored and restored

Restic saves snapshot e709d068, restores it separately, and diff confirms the recovered archive is identical to the transferred original.

```
labadmin@HQ-BACKUP:~$ sudo mountpoint -q /srv/backups && sudo restic -r /srv/backups/primecore backup /home/labadmin/backup-staging/hq-dc --tag hq-dc-ad
[sudo: authenticate] Password:
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
no parent snapshot found, will read all files
[0:00] 100.00% 1 / 1 index files loaded

Files:          1 new,      0 changed,      0 unmodified
Dirs:           4 new,      0 changed,      0 unmodified
Added to the repository: 1.396 MiB (1.360 MiB stored)

processed 1 files, 1.394 MiB in 0:07
snapshot e709d068 saved
labadmin@HQ-BACKUP:~$ sudo restic -r /srv/backups/primecore restore latest --tag hq-dc-ad --target /var/tmp/hq-dc-restore-check
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
[0:00] 100.00% 2 / 2 index files loaded
restoring snapshot e709d068 of [/home/labadmin/backup-staging/hq-dc] at 2026-09-30 04:42:30.016570392 +0000 UTC by root@HQ-BACKUP
P to /var/tmp/hq-dc-restore-check
Summary: Restored 5 files/dirs (1.394 MiB) in 0:00
labadmin@HQ-BACKUP:~$ sudo diff -s /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 /var/tmp/hq-dc-restore-check/home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2
Files /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 and /var/tmp/hq-dc-restore-check/home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 are identical
labadmin@HQ-BACKUP:~$ _
```

Figure B51 AD archive stored and restored